

EXPERIMENT 5

DPCM and Delta Modulation

Beginner-friendly theory + first-principles Python implementation + validation

Software-Based Digital Communication Laboratory

Python / Google Colab

1. Experiment Objectives

Taken directly from the uploaded laboratory manual, page 8.

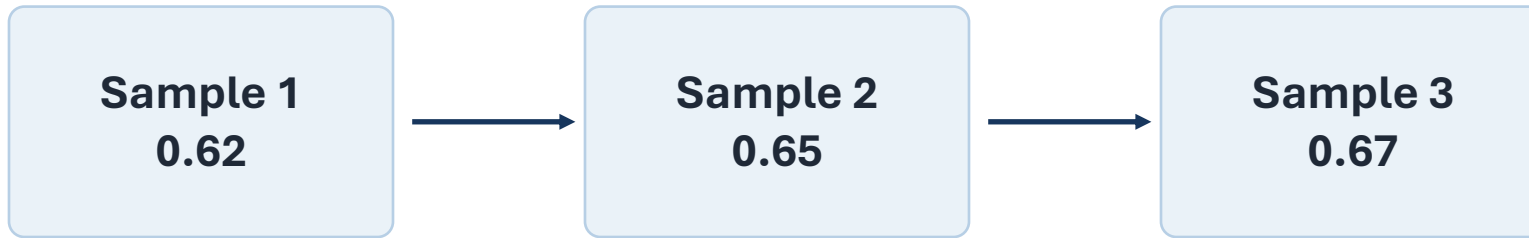
- Use prediction to reduce sample redundancy.
- Identify granular noise and slope-overload distortion.
- Implement a first-order DPCM predictor and compare it with PCM.
- Implement delta modulation for small, moderate and large step sizes.
- Test slowly and rapidly varying inputs.

Required visuals

Original vs predicted samples • Prediction error • Delta staircase • MSE vs step size

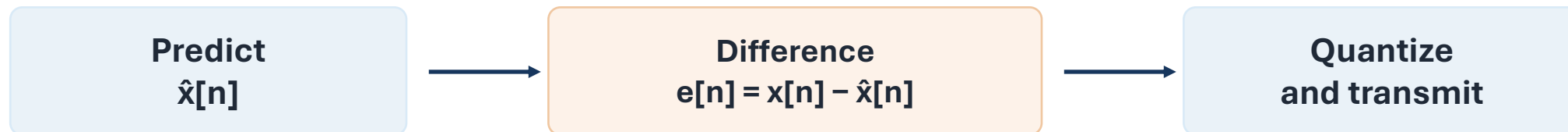
2. What Problem Are We Solving?

Start from the intuition before the equations.



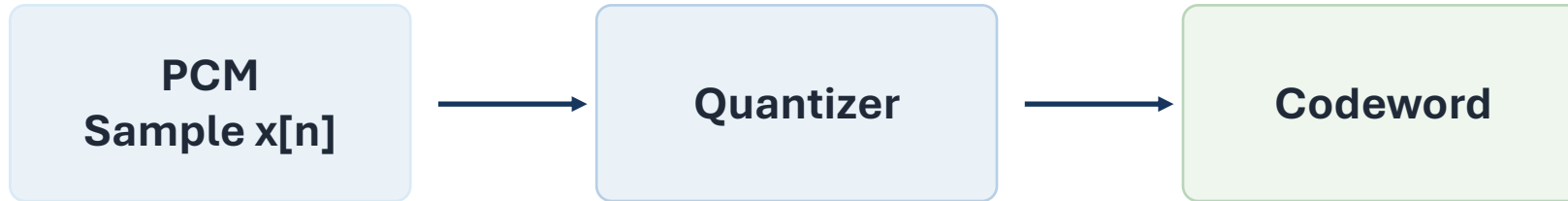
Nearby samples are often similar → there is redundancy.

DPCM sends the CHANGE from a prediction rather than sending every sample from scratch.



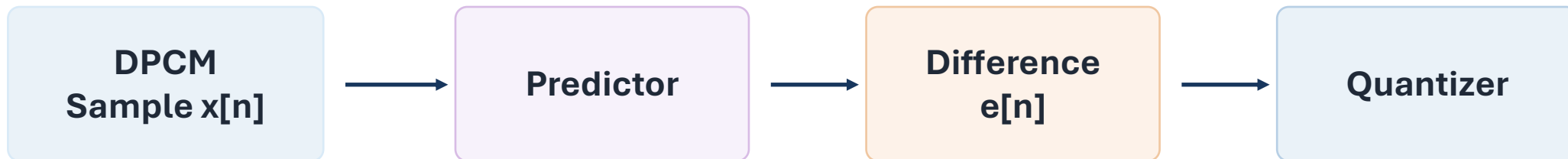
3. PCM vs DPCM

The difference is what gets quantized.



PCM path

$x[n] \rightarrow \text{quantizer} \rightarrow \text{codeword}$



DPCM reduces the amplitude range that the quantizer usually has to represent.

4. First-Order Prediction

For this experiment, the simplest predictor uses the previous reconstructed sample.

Predictor equation

$$\hat{x}[n] = \hat{x}[n-1]$$

Prediction error

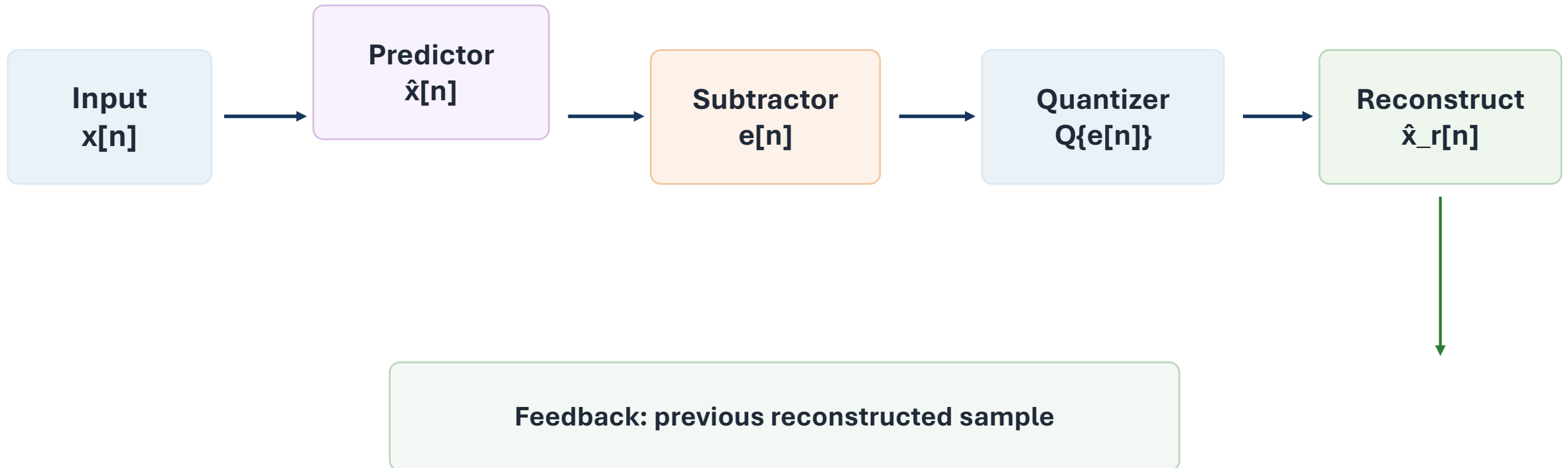
$$e[n] = x[n] - \hat{x}[n]$$

Why it helps

- For a smooth waveform, consecutive samples are close.
- The prediction error has a smaller dynamic range than the original signal.
- A fixed number of bits can therefore represent the error more finely.

Practical detail: the encoder and decoder must use the same predictor rule.

5. DPCM Transmitter–Receiver Chain



Decoder performs the same reconstruction rule using the quantized error.

6. DPCM Algorithm

A simple loop is enough to implement the experiment from first principles.

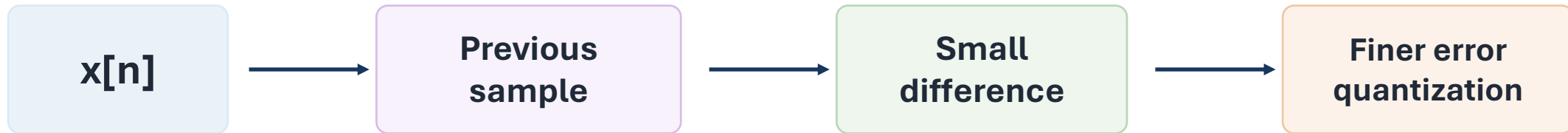
```
for n = 1, 2, ..., N-1:  
    prediction = reconstructed[n-1]  
    error = input[n] - prediction  
    quantized_error = Q(error)  
    reconstructed[n] = prediction + quantized_error
```



The loop makes the prediction state visible and reproducible.

7. Why a Smooth Signal Helps DPCM

Example used in the simulation: a 5 Hz sinusoid sampled at 1000 Hz.

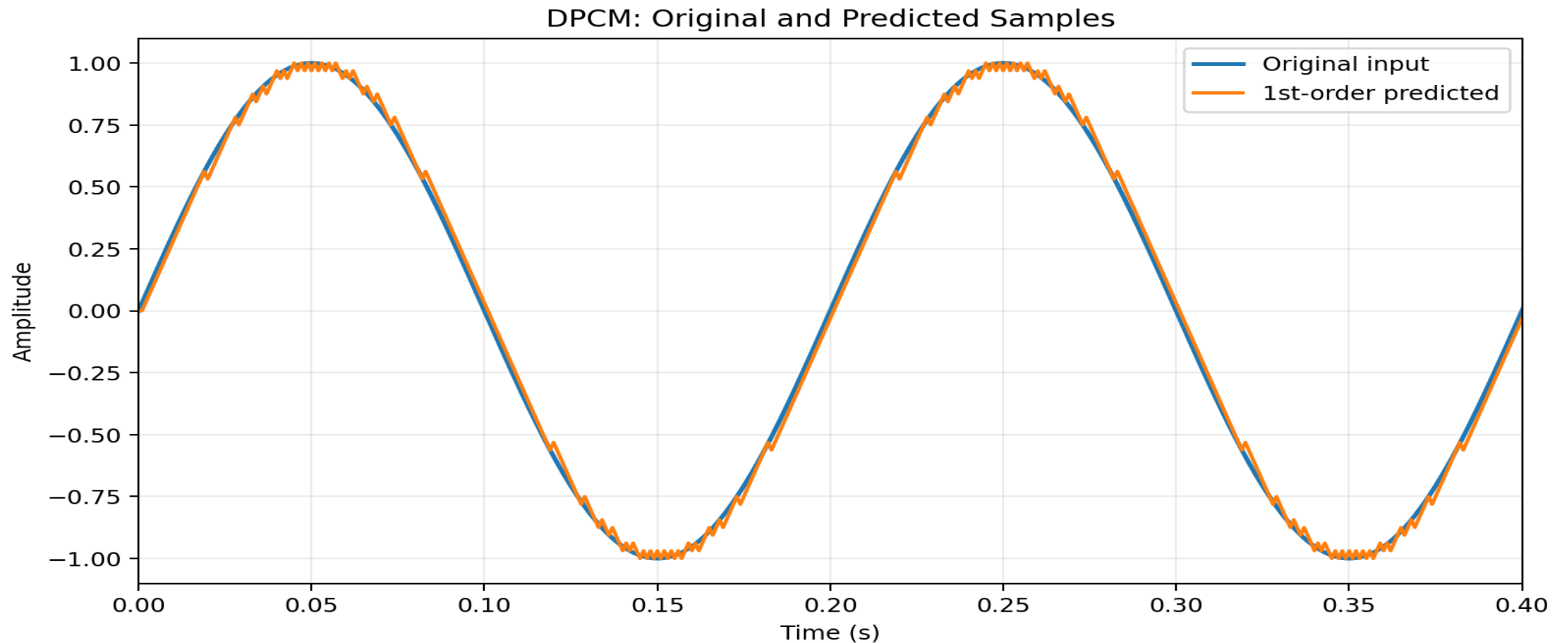


For a slow sinusoid, neighboring samples differ only slightly.

- The predictor follows the waveform closely.
- Prediction error is concentrated near zero.
- That leaves more effective quantization resolution around the typical error values.

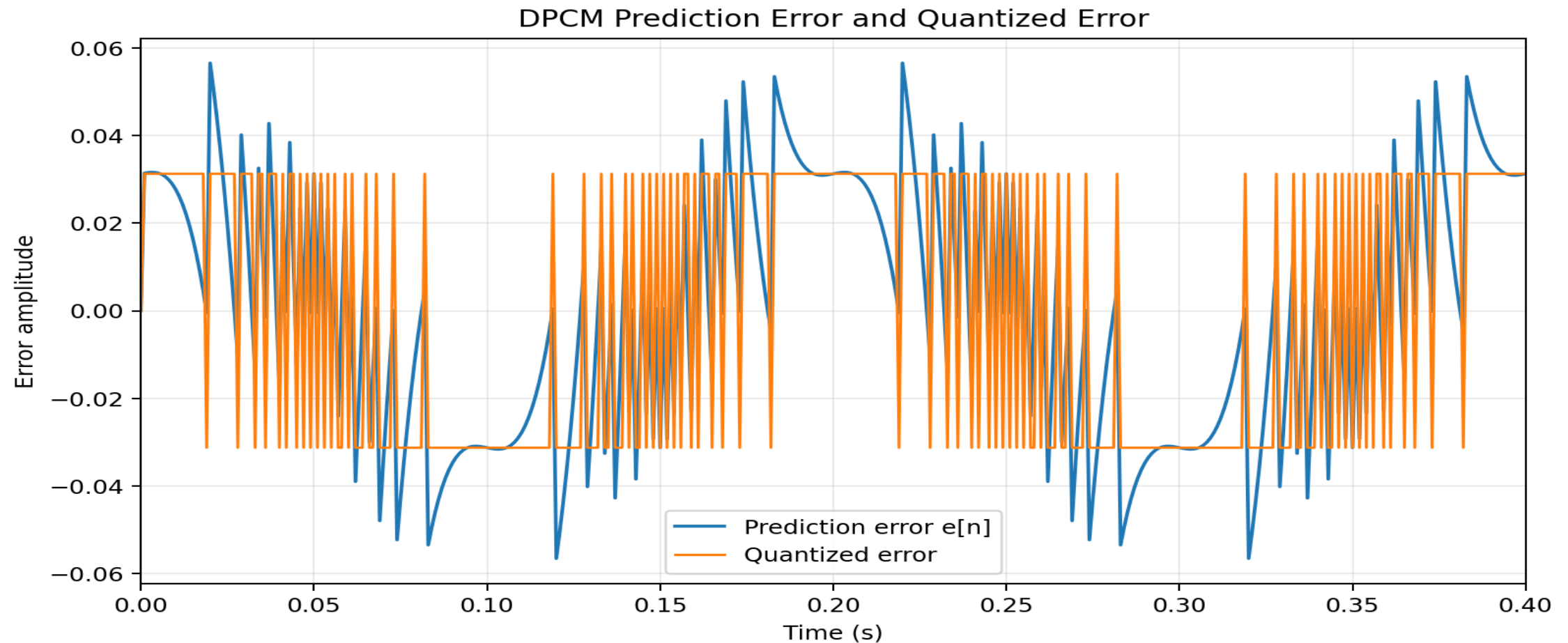
This is the redundancy-reduction idea behind DPCM.

8. Result — Original and Predicted Samples



The one-sample predictor closely follows the slow sinusoid, so the difference signal remains small.

9. Result — DPCM Prediction Error



Prediction error is the quantity that DPCM actually quantizes.

10. PCM vs DPCM — Quantization Error

Same 4-bit comparison on the 5 Hz input.

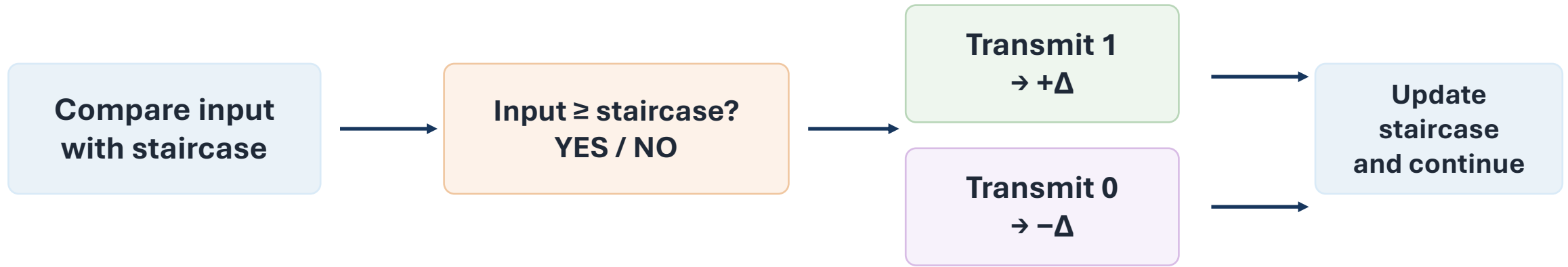
Method	MSE	What is quantized
4-bit PCM	0.001501	Original sample $x[n]$
4-bit DPCM	0.000271	Prediction error $e[n]$

Observed in this implementation

- DPCM has a lower MSE for this smooth input and chosen error range.
- The exact numerical advantage depends on predictor, bit depth and quantizer range.

11. Delta Modulation — The Core Idea

A one-bit differential quantizer: only the direction of change is transmitted.



Exactly one bit per sample: UP or DOWN.

This simplicity creates the two key distortions studied in the experiment.

12. Delta Modulation Equations

$$s[n] = s[n-1] + \Delta, \text{ when } x[n] \geq s[n-1]$$

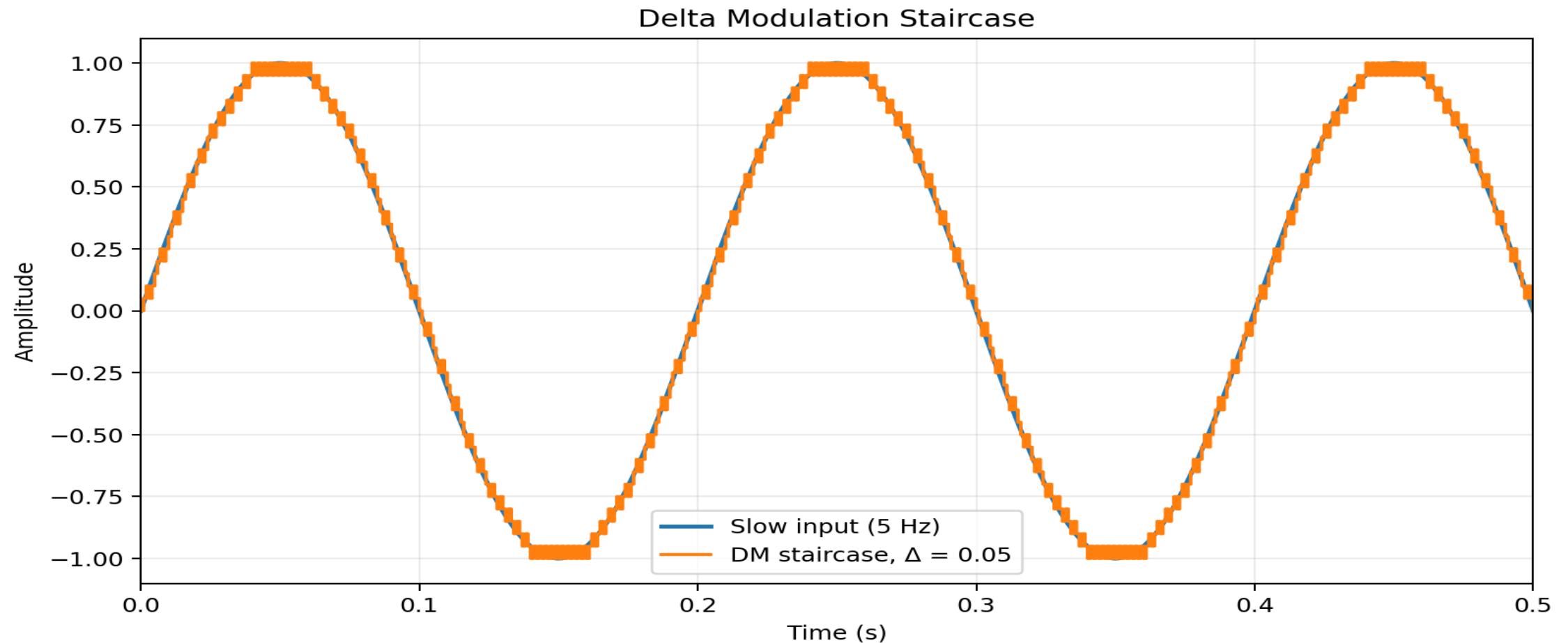
$$s[n] = s[n-1] - \Delta, \text{ when } x[n] < s[n-1]$$

Where

- $s[n]$ is the reconstructed staircase.
- Δ is the fixed step size.
- Every valid update changes the staircase by exactly $+\Delta$ or $-\Delta$.

This exact $\pm\Delta$ property is the manual's mandatory validation.

13. Result — Delta Modulation Staircase



For a moderate Δ , the staircase follows the slow input with bounded tracking error.

14. What Happens When Δ Changes?

Step size controls the maximum movement per sample.

Small Δ

**Tracks slowly
and precisely**

Risk: slope overload

Moderate Δ

**Balances tracking
and ripple**

Often practical

Large Δ

Jumps quickly

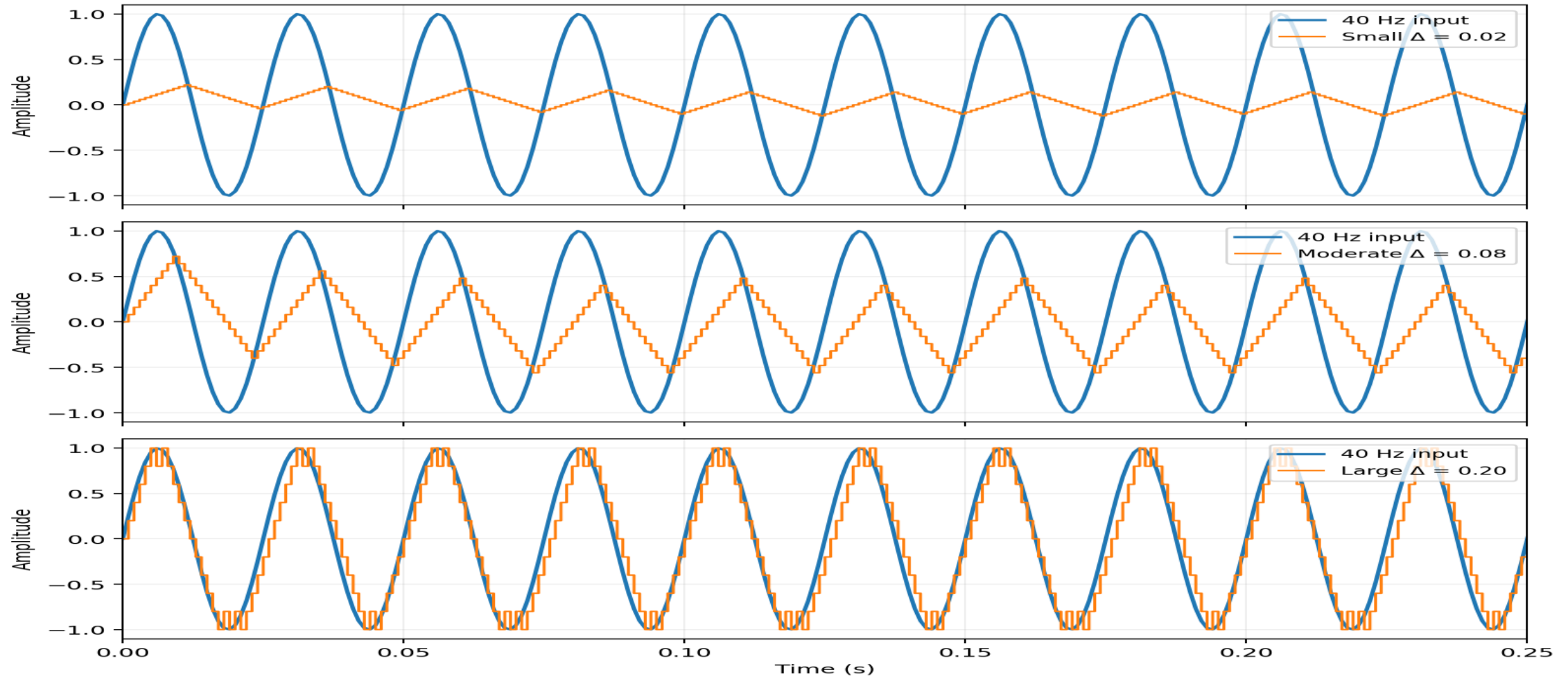
Risk: granular noise

There is no single step size that is ideal for every input.

Fast-changing signals demand a larger staircase slope; large steps then create more quantization-like ripple.

15. Small, Moderate and Large Δ on a Rapid Input

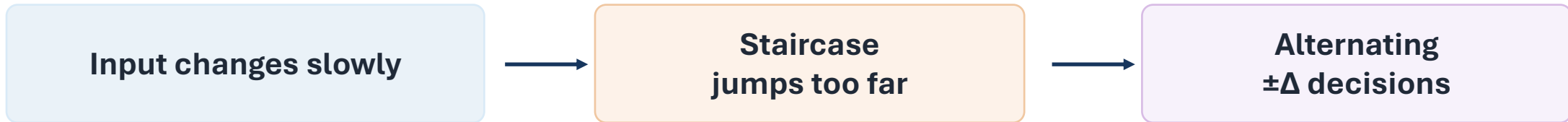
Delta Modulation: Effect of Step Size on a Rapid Input



On the 40 Hz input, a very small step visibly fails to keep up; a larger step improves tracking but introduces more ripple.

16. Granular Noise

Granular noise occurs when Δ is too large for the local signal variation.



- Large steps create a noticeable ripple around the desired waveform.
- Reducing Δ reduces this ripple—until the staircase becomes too slow for the input.
- Granular noise is therefore a small-step-versus-large-step trade-off, depending on the signal.

Think: “Too big to be precise.”

17. Slope Overload Distortion

Slope overload occurs when the input changes faster than the staircase can move.

Condition for a sinusoid $m(t) = A \sin(2\pi f_m t)$

$$|dm/dt|_{\max} = 2\pi f_m A$$

$$\text{Maximum DM slope} \approx \Delta/T_s$$

No-slope-overload requirement

$$2\pi f_m A \leq \Delta/T_s$$

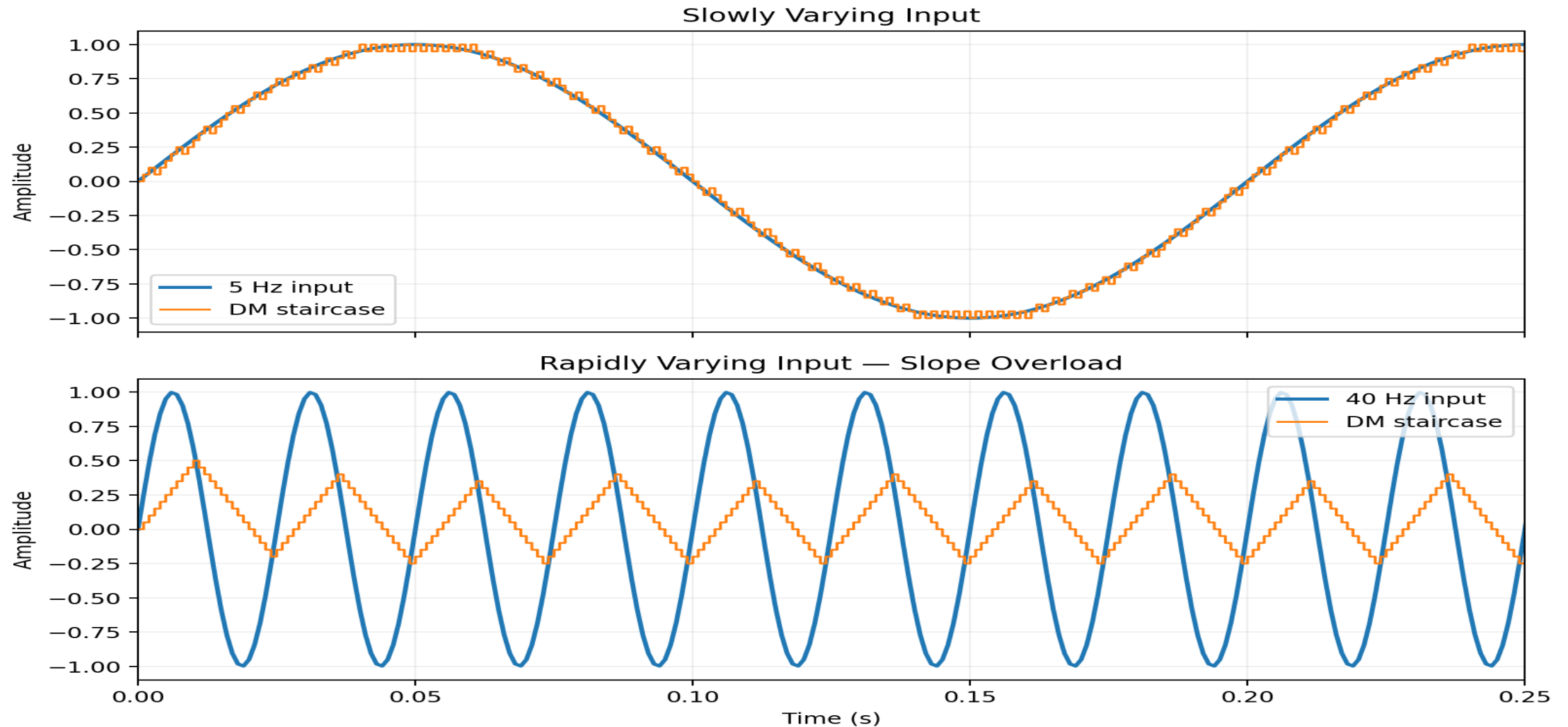
Simulation example: $f_s = 1000 \text{ Hz}$, $A = 1$, $\Delta = 0.05$

5 Hz input $\rightarrow$ max slope ≈ 31.4
40 Hz input $\rightarrow$ max slope ≈ 251.3
DM slope = 50.0 amplitude/s

5 Hz: below the DM slope limit

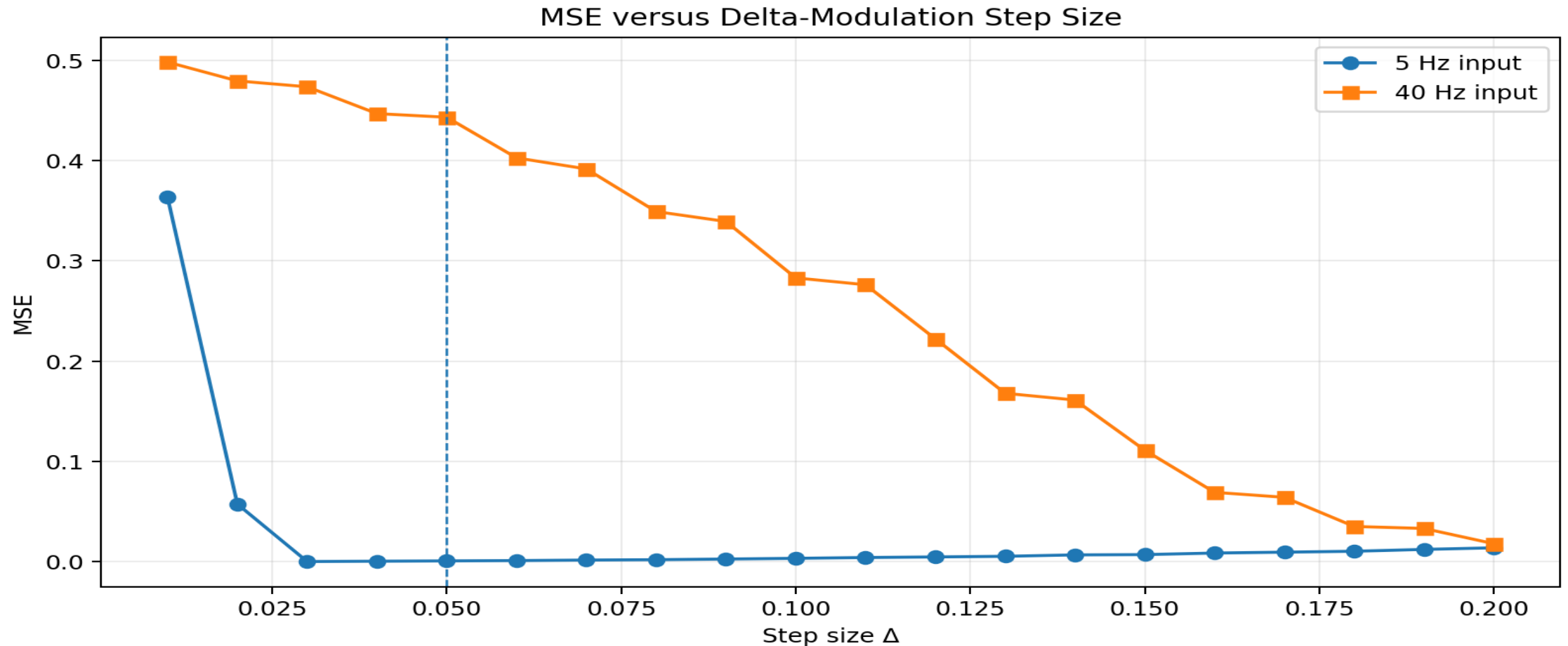
40 Hz: far above the DM slope limit

18. Result — Slow vs Rapid Inputs



Same $\Delta = 0.05$, different input frequency $\rightarrow$ different tracking behavior.

19. Result — MSE Versus Step Size



- For the slow input, very small Δ can increase tracking error because the staircase moves slowly.
- For the rapid input, too-small Δ produces strong slope-overload error; increasing Δ can improve tracking over part of the range.

20. Mandatory Validation Checks

The manual requires numerical validation, not only visual inspection.

Check	Simulation setting	Observed result
DPCM	4-bit, 5 Hz input	Closed-loop predictor reconstructs the waveform from quantized error
DM step rule	$\Delta = 0.05$	Every non-zero staircase step has magnitude 0.05
Slope test	$f_s=1000\text{ Hz}$, $\Delta=0.05$	5 Hz is below slope limit; 40 Hz is above it

DM assertion passed: all valid output steps are exactly $\pm\Delta$.

21. Python Setup — Parameters

```
import numpy as np
import matplotlib.pyplot as plt

fs = 1000
t = np.arange(0, 1.0, 1/fs)
slow = np.sin(2*np.pi*5*t)
fast = np.sin(2*np.pi*40*t)
```

Sampling rate
fs = 1000 Hz

Slow test
f = 5 Hz

Rapid test
f = 40 Hz

Using the same sampling grid makes waveform and step-size comparisons consistent.

22. Python Core Functions — DPCM and Delta Modulation

```
def dpcm_encode_decode(x, bits=4, error_range=0.5):
    recon = np.zeros(len(x))
    pred = np.zeros(len(x))
    err_q = np.zeros(len(x))
    recon[0] = x[0]
    for n in range(1, len(x)):
        pred[n] = recon[n-1]
        err = x[n] - pred[n]
        err_q[n] = quantize(err, bits, -error_range, error_range)
        recon[n] = pred[n] + err_q[n]
    return pred, recon

def delta_modulate(x, delta):
    staircase = np.zeros(len(x))
    bits = np.zeros(len(x), dtype=int)
    for n in range(1, len(x)):
        if x[n] >= staircase[n-1]:
            bits[n] = 1
            staircase[n] = staircase[n-1] + delta
        else:
            staircase[n] = staircase[n-1] - delta
    return bits, staircase
```

The full corrected program is supplied separately with the presentation.

23. Viva Questions You Should Be Ready For

What is DPCM?

A predictive coding method that quantizes the difference between a sample and its prediction.

Why does DPCM reduce redundancy? Neighboring samples are correlated, so the prediction error is often smaller than the original sample range.

What is delta modulation?

A 1-bit differential scheme that sends the direction of the next staircase step.

What is granular noise?

Tracking ripple caused by a step size that is too large for slowly varying input regions.

What is slope overload?

Tracking failure when the input slope is greater than the maximum staircase slope Δ/T_s .

How do you reduce slope overload? Increase Δ or increase the sampling rate, provided the design trade-offs are acceptable.

What is the mandatory DM check? Verify that each output staircase change is exactly $+\Delta$ or $-\Delta$.

24. Observation, Interpretation and Conclusion

Observation

- The predictor follows the slow input closely and reduces the typical error range.
- Delta modulation moves the reconstruction by fixed $\pm\Delta$ steps.
- Large Δ can create granular noise, while small Δ can cause slope overload on rapid inputs.

Interpretation

- DPCM exploits temporal correlation; its result depends on predictor and quantizer design.
- Delta modulation trades tracking ability against staircase ripple through a single parameter, Δ .

Final takeaway

DPCM encodes prediction error. Delta modulation encodes only UP/DOWN changes. Step size must be matched to signal variation.